Gauge Covariance and Geometric Refinement

Ordered transport, curvature, and the control of changing resolution

Transport beyond the free field

The free Dirac construction resolves a translation-invariant field into independent Fourier modes. Gauge coupling changes that organization. Transport between neighboring fibers carries additional information, and a comparison between resolutions must preserve the order and orientation of the links.

A link has a named source and target. Its transformation law is fixed by that direction. Composition cancels the gauge factors at internal endpoints, while a closed path retains conjugation at its basepoint. This elementary mechanism supports the treatment of holonomy, curvature, covariant differences, and refinement.

The finite Maxwell–Dirac example combines matter continuity, electric work, and propagation of the Gauss constraint. Geometric interpolation then separates edge compatibility, curvature defects, and Hilbert normalization. Regularity assumptions turn these structural conditions into quantitative estimates.

The constructions concern specified finite systems and prescribed background connections. A coupled continuum gauge theory additionally requires an operator domain, existence of the field evolution, propagation of constraints, and a convergence theorem in an explicitly chosen topology. Local covariance does not discharge those obligations.

The visual atlas develops the finite examples. The technical appendix records formal source coverage and the limits of its verification evidence. The argument here remains readable without following implementation identifiers.

Reading routes and conventions

Begin with endpoint covariance if gauge theory is new to you. The cancellation of intermediate frame changes explains both open-path covariance and closed-loop conjugation. In an Abelian group, conjugation disappears; in a non-Abelian group, the trace or conjugacy class is the natural loop observable. The ordered product always records which link acts first.

For finite dynamics, continue to the semidiscrete matter–electric-field system. Here continuity, preservation of a Gauss residual, and energy exchange follow from a specified finite system of equations. The distinction between a time-dependent frame change and the residual symmetry after choosing temporal gauge is made explicit. The consistency calculation also states how a smooth connection is converted into a link; its sampling rule affects the order of accuracy.

For refinement, follow the second part from actual half-edge geometry to cubical paths and Hilbert norms. The text keeps the straight-edge mismatch separate from the curvature contribution. Regular line-integral links remove the former and bound the latter. The reconstruction discussion then explains what is known about state spaces and which generator or evolution comparisons still have to be proved.

Convention	Meaning
Inner product	$\langle v, w\rangle$ is conjugate-linear in v and linear in w. Matrix adjoints are written T^*.
Endpoint transport	$T_{y\leftarrow x}$ maps the fiber at x to the fiber at y. Under a local unitary frame g, it becomes $g_y T_{y\leftarrow x} g_x^{-1}$.
Matrix order	$T_{z\leftarrow y}T_{y\leftarrow x}$ transports first from x to y, then from y to z. The rightmost factor acts first.
Linked shifts	A coefficient U_j in $U_j\psi_{j+1}$ transports from $j+1$ back to j. Thus $U'_j = g_j U_j g_{j+1}^{-1}$. An outward link uses the opposite endpoint convention.
Gauge groups	$U(1)$ links are unit complex numbers. The finite non-Abelian construction uses unitary matrices, with $SU(2)$ as an explicit example. Scalar multiplication identities are not silently generalized to noncommuting matrices.
Mesh and time	Mesh widths are positive when dividing by them. The finite matter system uses $\hbar = 1$ and the displayed charge convention. Refinement level and evolution time are different parameters.
Norm and measure	Spinor and operator norms are specified in the statement. Half-edge and cubical interpolation use their stated cell weights; an unweighted duplication map is not automatically isometric.
Background field	A prescribed connection supplies link integrals and curvature estimates. A dynamical gauge field additionally needs its own evolution and constraints.

The geometric source modules also use forward-link notation. The source notes translate it explicitly into the endpoint convention above. In particular, the word “first” in an edge-subdivision structure is read together with its product equation and endpoint convention.

Gauge transport and finite dynamics

1 Links compare neighboring fibers

A field assigns a vector to each site, but a local change of frame can be chosen independently at every site. A difference such as $\psi_{x+\widehat{\mu}} - \psi_x$ therefore needs a rule for comparing its two entries. A link supplies that rule. Its orientation is part of its definition, rather than a sign convention to be inferred after a calculation.

Let V be a finite periodic lattice and let every fiber be a copy of $\mathbb{C}^r$. We initially allow any unitary gauge group acting on this fiber. In the convention used for the spatial stencils below,

$$U_{x,\mu} : \mathcal{H}_{x+\widehat{\mu}} \longrightarrow \mathcal{H}_x, \qquad U_{x,\mu}^* U_{x,\mu} = U_{x,\mu} U_{x,\mu}^* = I. \tag{1}$$

The gauge $g = (g_x)_{x\in V}$ changes coordinates by $\psi_x^g = g_x \psi_x$. Requiring the transported vector to have the frame of its destination gives

$$U_{x,\mu}^g = g_x U_{x,\mu} g_{x+\widehat{\mu}}^{-1}, \qquad U_{x,\mu}^g \psi_{x+\widehat{\mu}}^g = g_x U_{x,\mu} \psi_{x+\widehat{\mu}}. \tag{2}$$

Products act on column vectors from right to left. More generally, a transporter $T_{y\leftarrow x}$ has the transformation $g_y T_{y\leftarrow x} g_x^{-1}$. This notation checks both the input and output of every product.

The forward link is

$$u_{x,\mu} = U_{x,\mu}^{-1} : \mathcal{H}_x \longrightarrow \mathcal{H}_{x+\widehat{\mu}}, \qquad u_{x,\mu}^g = g_{x+\widehat{\mu}} u_{x,\mu} g_x^{-1}. \tag{3}$$

The older time-dependent Abelian walk writes this forward link as L_j. The later geometric Lean sources use u. The non-Abelian stencil and the semidiscrete Hamiltonian use the backward U. These formulas describe the same edge in opposite directions. In particular, reversing an edge changes its transporter to the inverse, not to the same matrix in a different notation.

Proposition 1.1 (Covariance of linked differences)**.** *For spacing $h > 0$, define*

$$(D_{\mu,U}^+ \psi)_x = \frac{U_{x,\mu}\psi_{x+\widehat{\mu}} - \psi_x}{h}, \tag{4}$$

$$(\Delta_U \psi)_x = \frac{1}{h^2} \sum_\mu \left(U_{x,\mu}\psi_{x+\widehat{\mu}} + U_{x-\widehat{\mu},\mu}^{-1}\psi_{x-\widehat{\mu}} - 2\psi_x \right). \tag{5}$$

Then $D_{\mu,U^g}^+(g\psi) = g D_{\mu,U}^+ \psi$ and $\Delta_{U^g}(g\psi) = g\Delta_U\psi$.

Proof. For the positive edge, substitute (2). The two gauge factors at $x + \hat{\mu}$ cancel, leaving g_x on the left. For the negative edge,

$$(U_{x-\widehat{\mu},\mu}^g)^{-1} g_{x-\widehat{\mu}} = g_x U_{x-\widehat{\mu},\mu}^{-1}.$$

Every term in either difference therefore has the common factor g_x. □

The lattice inner product is $\langle \phi, \psi \rangle_V = \sum_x \phi_x^* \psi_x$, conjugate-linear in the first entry. Periodic reindexing and unitarity also give the useful identity

$$-\langle \psi, \Delta_U \psi \rangle_V = \sum_{x,\mu} \left\| (D_{\mu,U}^+ \psi)_x \right\|^2. \tag{6}$$

Indeed, expanding one squared difference gives $h^{-2}(\|\psi_{x+\widehat{\mu}}\|^2 + \|\psi_x\|^2 - 2\,\mathrm{Re}(\psi_x^* U_{x,\mu}\psi_{x+\widehat{\mu}}))$. Summing and shifting the first norm term produces the left side. The forward and backward hopping blocks are adjoints, so Δ_U is self-adjoint and nonpositive. This finite identity uses the periodic boundary condition; an open domain needs its boundary terms or a specified boundary condition.

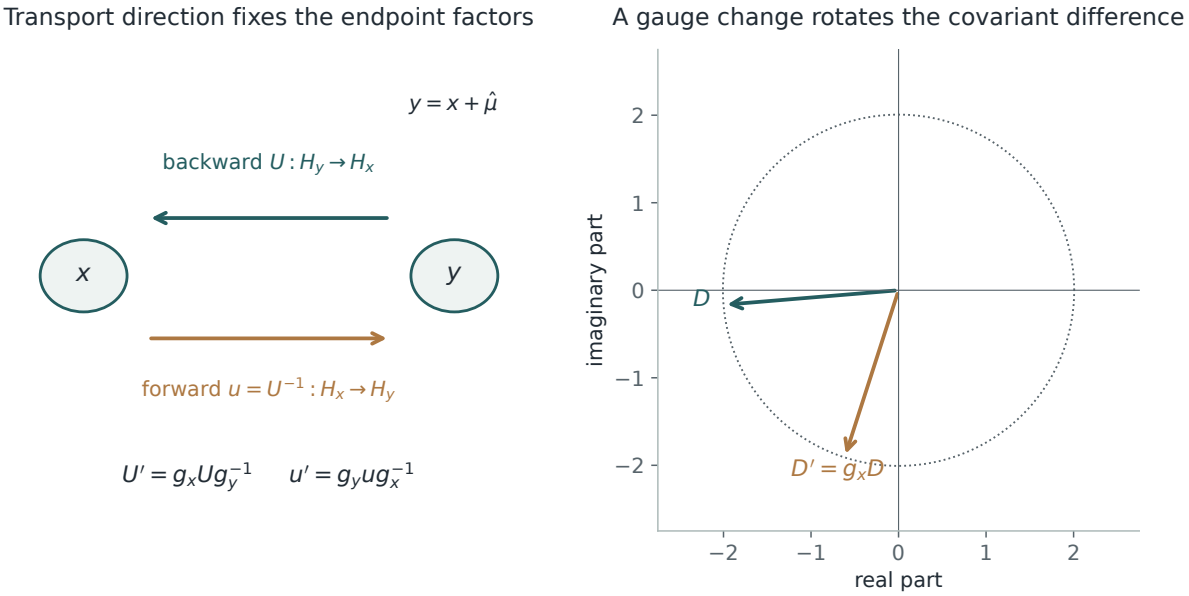


Figure 1. A backward link U and its forward inverse u compare the same two fibers. Gauge factors cancel at the input and remain at the destination. The scalar example illustrates the exact transformation law.

2 A linked walk and its actual charge flux

Covariance alone does not specify a current. To obtain one, fix the walk that moves the amplitudes. Let $N \geq 1$ be an integer, let $j \in \mathbb{Z}/N\mathbb{Z}$, let $\psi_j^n \in \mathbb{C}^2$, and let $U_j^n \in U(1)$ be backward spatial links. Let $T_j^n \in U(1)$ transport from (j, n) to $(j, n+1)$. At each step choose two on-site unitary coins $C_{1,j}^n, C_{2,j}^n \in U(2)$ and define

$$\eta_j = C_{1,j}^n \psi_j^n, \tag{7}$$

$$\xi_j^+ = (U_{j-1}^n)^{-1}\eta_{j-1}^+, \qquad \xi_j^- = U_j^n \eta_{j+1}^-, \tag{8}$$

$$\psi_j^{n+1} = T_j^n C_{2,j}^n \xi_j. \tag{9}$$

Here the $+$ channel moves toward increasing j, while the $-$ channel moves toward decreasing j. The legacy implementation takes the two coins to be equal half mass rotations. The argument needs only their unitarity and the fact that scalar phases commute with them.

For a time-dependent scalar gauge g_j^n, use

$$(U_j^n)^g = g_j^n U_j^n (g_{j+1}^n)^{-1}, \qquad (T_j^n)^g = g_j^{n+1} T_j^n (g_j^n)^{-1}. \tag{10}$$

Spatial transport first brings the gauge factor to site j on the old slice. Temporal transport then brings it to the new slice. Thus the step operator $\mathcal{W}_n$ satisfies

$$\mathcal{W}_n[U^g, T^g]G^n = G^{n+1}\mathcal{W}_n[U, T], \qquad (G^n\psi)_j = g_j^n\psi_j. \tag{11}$$

Every factor in the step is unitary on the finite state space: the two coins and temporal phases act sitewise, and the linked shift is a permutation of the separate channels followed by unit-modulus multiplication.

Proposition 2.1 (Exact one-step continuity)**.** *Set* $\rho_j^n = \|\psi_j^n\|^2$ *and*

$$J_{j+1/2}^n = |\eta_j^+|^2 - |\eta_{j+1}^-|^2. \tag{12}$$

The linked walk obeys

$$\rho_j^{n+1} - \rho_j^n = J_{j-1/2}^n - J_{j+1/2}^n. \tag{13}$$

Both the density and this current are gauge invariant. Total charge is conserved. For a consecutive periodic arc A *with specified left and right boundary edges,* $Q_A^{n+1} - Q_A^n = J_{\partial_- A}^n - J_{\partial_+ A}^n$.

Proof. The first coin preserves the local norm, so $\rho_j^n = |\eta_j^+|^2 + |\eta_j^-|^2$. The second coin and temporal phase preserve the norm after the shift, giving $\rho_j^{n+1} = |\eta_{j-1}^+|^2 + |\eta_{j+1}^-|^2$. Subtract these expressions and group the two terms associated with each boundary edge. This is (13). The transformed η_j is $g_j^n\eta_j$, so its channel norms are unchanged. Summing the balance over the full cycle or the chosen arc cancels its internal currents. □

This current measures charge transferred during one discrete step. Dividing it by a chosen time step would give a rate. It is not the same expression as the instantaneous Hamiltonian current introduced below. The two expressions belong to different evolution laws, even though both produce a local balance by cancellation.

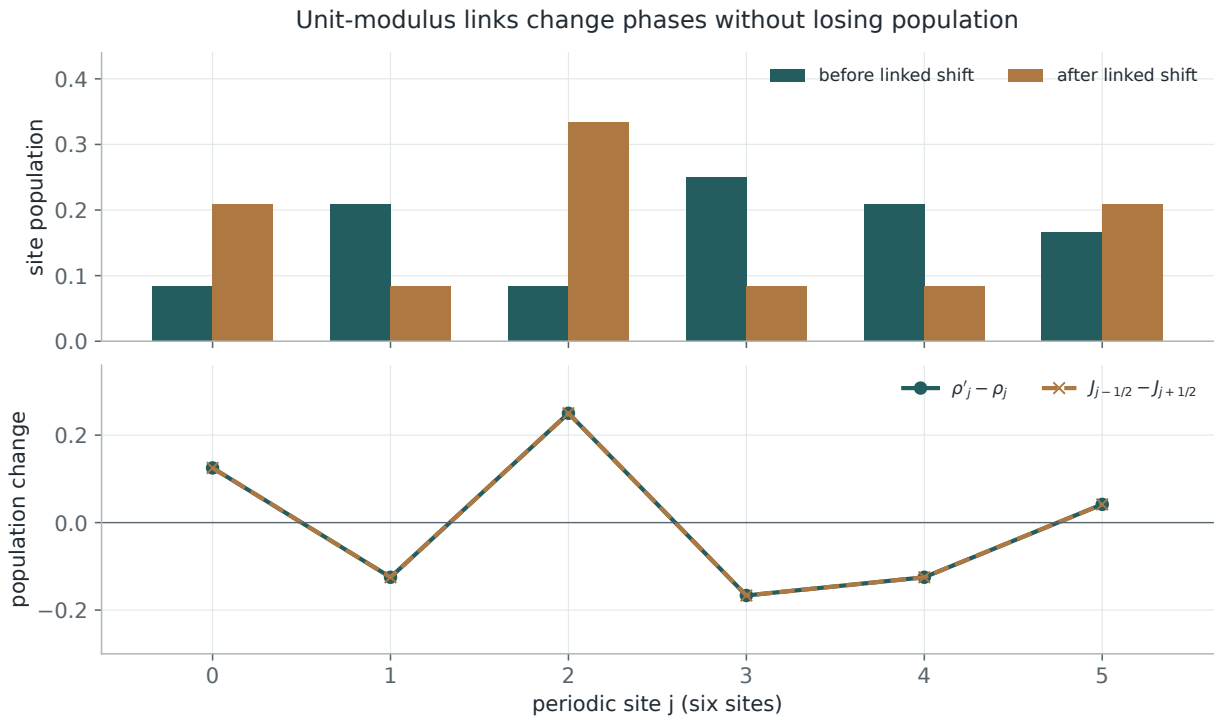


Figure 2. The explicit finite coined walk supplies a current before a continuum interpretation is attempted. The plotted computation checks one-step local balance and total population for its declared finite data.

3 Unit-circle Clifford channels in the formal transport layer

The later formal development makes the directional channels depend on the Clifford matrix of that direction. Let $A = A^*$ and $A^2 = I$ on a finite spinor space. Then

$$P_A^+ = \tfrac{1}{2}(I + A), \qquad P_A^- = \tfrac{1}{2}(I - A) \tag{14}$$

are self-adjoint complementary orthogonal projections. Idempotence and $P_A^+ P_A^- = 0$ follow by expanding the products and using $A^2 = I$; self-adjointness follows from $A = A^*$. Different spatial Clifford matrices generally have different spectral projections. A permanent assignment of the four spinor coordinates to two Boolean channels cannot replace this construction in all three directions.

For a bijection $\tau : V \to V$ and a forward scalar link $u : V \to U(1)$, define

$$(S_{A,\tau,u}\psi)(x) = u(\tau^{-1}x)P_A^+\psi(\tau^{-1}x) + u(x)^{-1}P_A^-\psi(\tau x). \tag{15}$$

This agrees with the orientation dictionary (3). Its inverse is

$$(S_{A,\tau,u}^{-1}\phi)(x) = u(x)^{-1}P_A^+\phi(\tau x) + u(\tau^{-1}x)P_A^-\phi(\tau^{-1}x). \tag{16}$$

Substitution proves both inverse identities: same-channel terms cancel the link with its inverse and mixed-channel terms vanish. Orthogonality, followed by reindexing x through τ in each channel, gives

$$\sum_x \langle S\psi(x), S\phi(x)\rangle = \sum_x \langle \psi(x), \phi(x)\rangle\,. \tag{17}$$

For $u^g(x) = g(\tau x)u(x)g(x)^{-1}$, direct endpoint cancellation gives $S_{A,\tau,u^g}G_g = G_g S_{A,\tau,u}$. Products of these directional steps and scalar-gauge-equivariant unitary mass steps retain covariance and inner-product preservation.

The genuine unit-circle hypothesis is essential. A nonzero complex scalar is invertible, but multiplication by 2 multiplies squared norm by 4. The phase must therefore lie in the unit circle, not merely in $\mathbb{C}^\times$. Norm preservation depends on that mathematical hypothesis. The formalization companion identifies its precise type and transport conventions.

The checked four-spinor construction in this layer uses scalar $U(1)$ phases. The $SU(2)$ matrices of the legacy non-Abelian chapter act on an internal two-dimensional fiber. To combine that color action with a Dirac spinor, one would specify a space such as $\mathbb{C}^4 \otimes \mathbb{C}^2$, with spin projectors on the first factor and gauge transport on the second. This guarantees their commutation. No assertion here identifies the two-dimensional color fiber with the four-dimensional Clifford spinor or declares the resulting non-Abelian Dirac extension checked.

4 Plaquettes, invariant observables, and curvature phases

On a two-dimensional coordinate plane, the backward links define the based plaquette

$$W_{x,\mu\nu} = U_{x,\mu}U_{x+\hat{\mu},\nu}U^{-1}_{x+\hat{\nu},\mu}U^{-1}_{x,\nu}. \tag{18}$$

Starting with a vector at x, the rightmost factor moves it to $x+\hat{\nu}$; the next moves it to the far corner; the last two bring it back to x. The matrix product is therefore an endomorphism of the base fiber. Each intermediate gauge factor cancels in its written order, giving

$$W^g_{x,\mu\nu} = g_x W_{x,\mu\nu} g_x^{-1}. \tag{19}$$

For $U(1)$ the plaquette itself is invariant. For a noncommutative group it is its conjugacy class that is invariant; the matrix entries depend on the frame at x.

For $SU(2)$ links, the Wilson action of the source is the finite function

$$S_W[U] = \frac{1}{g^2_{\mathrm{YM}}} \sum_{x,\mu<\nu} \left(1 - \tfrac{1}{2}\operatorname{Re}\operatorname{Tr} W_{x,\mu\nu}\right), \qquad g_{\mathrm{YM}} \in \mathbb{R}\setminus\{0\}. \tag{20}$$

The subscript distinguishes the coupling from the site gauge g_x. Cyclicity of trace proves $S_W[U^g] = S_W[U]$. Since an $SU(2)$ matrix has eigenvalues $e^{i\theta}, e^{-i\theta}$, a plaquette term is $1-\cos\theta \geq 0$. In the Abelian case the analogous term is $1 - \operatorname{Re} W$. Specifying an invariant action is not yet a construction of its Hamiltonian phase space, Gauss constraint, or a continuum interacting field theory.

4.1 An Abelian surface product

For scalar links, multiply the oriented plaquettes on the six faces of a cube. Every edge occurs twice, with opposite orientation. Commutativity permits its factors to meet and cancel, so

$$\prod_{f\subset\partial c} W_f^{\mathrm{or}(f,c)} = 1. \tag{21}$$

The same cancellation makes the product of fine plaquettes on a rectangular patch equal its boundary holonomy. The statement is an exact group identity. If $W_f = e^{i\theta_f}$ with each θ_f chosen as a principal

argument, it implies that the oriented phase sum lies in $2\pi\mathbb{Z}$. It does not force that sum to be zero as a real number. An unwrapped small-phase regime or a specified branch is needed for that stronger conclusion.

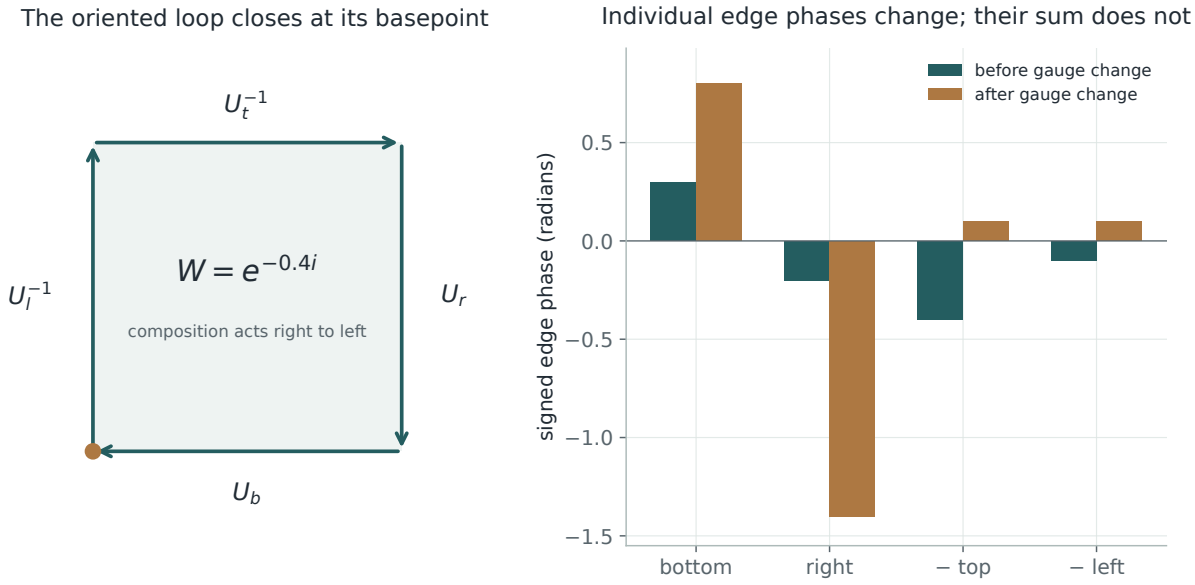


Figure 3. For Abelian links, the oriented edge phases cancel around adjacent faces. A holonomy is branch independent; a displayed real phase requires a branch choice.

4.2 Small smooth plaquettes

Suppose a smooth Abelian connection is sampled by exact line integrals, with a consistent loop orientation. Write its holonomy as $W = e^{i\theta}$, where the small real phase is $\theta = -q\int_{\square} F_{\mu\nu}\, dS$. Here q is the charge and $F_{\mu\nu}$ is the corresponding oriented curvature component. Take rectangles with side lengths $a_\mu, a_\nu = O(a)$ and uniformly controlled derivatives of F. If the rectangle is centered at x_c and F is C^2, symmetric integration removes its linear Taylor terms:

$$\theta = -q a_\mu a_\nu F_{\mu\nu}(x_c) + O(a^4). \tag{22}$$

Consequently,

$$1 - \operatorname{Re} W = \tfrac{1}{2} q^2 a_\mu^2 a_\nu^2 F_{\mu\nu}(x_c)^2 + O(a^6). \tag{23}$$

To see the remainder, use $1-\cos\theta = \theta^2/2 + O(\theta^4)$ and $\theta = O(a^2)$. The cross term from the $O(a^4)$ phase error is $O(a^6)$, while the fourth-power term is $O(a^8)$. Evaluating the curvature at a corner instead generally leaves an $O(a^3)$ phase error and hence an $O(a^5)$ action-density error. These orders concern the chosen pointwise comparison and exact-integral sampling. They do not by themselves justify convergence of an action sum or its solutions.

5 A finite Maxwell–Dirac model and its conserved constraint

The finite lattice can also carry dynamical link variables. The following is the particular semidiscrete model in the Abelian source, with time continuous and space a periodic cycle of $N \geq 3$ sites. It is useful because its matter current, electric work, and Gauss propagation can all be checked in the same equations.

Fix $h > 0$, $m \in \mathbb{R}$, and Hermitian spin matrices $\alpha = \alpha^*$ and $\beta = \beta^*$ on $\mathbb{C}^s$. Let $\psi_j \in \mathbb{C}^s$ and $A_j, E_j \in \mathbb{R}$, and set $U_j = e^{-ihA_j}$. The source absorbs the charge into A. Its real Hamiltonian is

$$H_h = H_{\mathrm{m}} + H_E, \tag{24}$$

$$H_{\mathrm{m}} = hm \sum_j \psi_j^* \beta \psi_j + \sum_j \mathrm{Im}(\psi_j^* \alpha U_j \psi_{j+1}), \qquad H_E = \frac{h}{2} \sum_j E_j^2. \tag{25}$$

Hermiticity is enough for the identities below. Additional Clifford relations specify a Dirac representation but are not needed to prove charge or energy conservation in this finite model. The action fixes all normalization and sign conventions:

$$\mathcal{S}_h = \int \left[\frac{ih}{2} \sum_j (\psi_j^* \dot{\psi}_j - \dot{\psi}_j^* \psi_j) + h \sum_j E_j \dot{A}_j - H_h \right] dt. \tag{26}$$

Variations with fixed endpoint data yield $ih\dot{\psi}_j = \partial H_h / \partial \overline{\psi}_j$, $h\dot{A}_j = \partial H_h / \partial E_j$, and $h\dot{E}_j = -\partial H_h / \partial A_j$. Written out, these are

$$\dot{\psi}_j = -im\beta\psi_j - \frac{1}{2h} \alpha U_j \psi_{j+1} + \frac{1}{2h} U_{j-1}^* \alpha \psi_{j-1}, \tag{27}$$

$$\dot{A}_j = E_j, \qquad \dot{E}_j = J_j, \qquad J_j = \mathrm{Re}(\psi_j^* \alpha U_j \psi_{j+1}). \tag{28}$$

For example, if $z_j = \psi_j^* \alpha U_j \psi_{j+1}$, then $\partial_{A_j} \operatorname{Im} z_j = \mathrm{Im}(-ihz_j) = -h \operatorname{Re} z_j$. This fixes the positive sign of $\dot{E}_j = J_j$ for the conventions in the action.

Theorem 5.1 (Charge balance and Gauss propagation)**.** *For every differentiable solution of* (27)–(28)*, define*

$$\rho_j = \psi_j^* \psi_j, \qquad \bar{\rho} = \frac{1}{N} \sum_j \rho_j, \qquad \mathcal{G}_j = \frac{E_j - E_{j-1}}{h} + \rho_j - \bar{\rho}. \tag{29}$$

Then

$$\dot{\rho}_j + \frac{J_j - J_{j-1}}{h} = 0, \qquad \dot{\bar{\rho}} = 0, \qquad \dot{\mathcal{G}}_j = 0. \tag{30}$$

In particular, initial data satisfying $\mathcal{G}_j = 0$ retain that constraint on their interval of existence.

Proof. Differentiate ρ_j and use $\dot{\rho}_j = 2\,\mathrm{Re}(\psi_j^* \dot{\psi}_j)$. The mass term contributes zero, since $\psi_j^* \beta \psi_j$ is real. The forward hopping term contributes $-J_j/h$. The backward scalar is the conjugate of $\psi_{j-1}^* \alpha U_{j-1} \psi_j$, using $\alpha = \alpha^*$ and the scalar nature of U. Its contribution is J_{j-1}/h. This proves continuity. Periodic summation gives $\dot{\bar{\rho}} = 0$. Finally,

$$\dot{\mathcal{G}}_j = \frac{\dot{E}_j - \dot{E}_{j-1}}{h} + \dot{\rho}_j - \dot{\bar{\rho}} = \frac{J_j - J_{j-1}}{h} - \frac{J_j - J_{j-1}}{h} = 0.$$

□

The subtraction of $\bar{\rho}$ is structural on a periodic cycle: the discrete divergence sums to zero, so a constraint containing a nonzero total charge without compensating background would be inconsistent. Equation (29) supplies that neutralizing mean. Conservation of $\mathcal{G}_j$ is a propagation theorem. It preserves an initially nonzero constraint violation too; it does not project arbitrary initial data onto the constraint surface. Summing continuity over $a \le j \le b$ gives the physical-length-weighted charge balance

$$\frac{d}{dt}\left(h\sum_{j=a}^{b}\rho_j\right) = J_{a-1} - J_b.$$

Proposition 5.2 (Electric work and total energy)**.** *Along the same solutions,*

$$\dot{H}_E = h\sum_j E_j J_j, \qquad \dot{H}_{\mathrm{m}} = -h\sum_j E_j J_j, \qquad \dot{H}_h = 0. \tag{31}$$

Proof. The first formula follows from $\dot{E}_j = J_j$. For the matter contribution, let $q_j = \partial H_{\mathrm{m}}/\partial\overline{\psi}_j = ih\dot{\psi}_j$. The state-dependent part of the real derivative is $2\,\mathrm{Re}\sum_j q_j^*\dot{\psi}_j = 0$, because $q_j^*\dot{\psi}_j = -ih\left\|\dot{\psi}_j\right\|^2$ is purely imaginary. The remaining explicit link derivative is

$$\sum_j \frac{\partial H_{\mathrm{m}}}{\partial A_j}\dot{A}_j = -h\sum_j J_j E_j.$$

Adding the two sectors proves total conservation. □

5.1 Which gauge transformations remain in this action?

For time-independent real χ_j, the transformation

$$\psi_j \mapsto e^{i\chi_j}\psi_j, \qquad A_j \mapsto A_j + \frac{\chi_{j+1}-\chi_j}{h}, \qquad E_j \mapsto E_j \tag{32}$$

leaves the action and equations invariant. In fact U_j then has precisely the backward-link law (2), every hopping scalar z_j is unchanged, and the time derivatives of the gauge parameters vanish.

This is the residual symmetry of the stated temporal-gauge model. If χ_j depends on time, the matter time derivative acquires $i\dot{\chi}_j\psi_j$, while $\dot{A}_j$ acquires $(\dot{\chi}_{j+1}-\dot{\chi}_j)/h$. With E_j fixed and no scalar potential in (26), these changes do not preserve the displayed equations. The fully time-dependent covariance of the discrete linked walk uses its temporal link T; it must not be transferred to this action after that degree of freedom has been removed.

The statements in this section are ordinary proofs for a declared finite-dimensional classical field model. They establish genuine interaction between its matter and electric variables. They do not constitute a Lean construction of a coupled gauge Hamiltonian, a quantized gauge field, or a global continuum Maxwell–Dirac solution.

6 What a second-order covariant stencil requires

The exact finite identities did not assume a smooth field in the background. A continuum comparison does. Let A be a real C^2 function near x, let ψ be C^3 there, and set

$$U_h(x) = \exp\left(-i\int_x^{x+h} A(s)\,ds\right), \qquad (D_h^A\psi)(x) = \frac{U_h(x)\psi(x+h) - U_h(x-h)^{-1}\psi(x-h)}{2h}. \tag{33}$$

The smooth A in this paragraph is a continuum coefficient. In the finite Hamiltonian one obtains the same link by taking A_j to be its edge average $h^{-1}\int_{x_j}^{x_j+h} A(s)\,ds$. This is an additional sampling prescription.

Proposition 6.1 (Local second-order consistency)**.** *With uniform bounds on A and its derivatives through order two and on ψ and its derivatives through order three on a fixed neighborhood,*

$$(D_h^A\psi)(x) = (\partial_x - iA(x))\psi(x) + O(h^2) \qquad (h\to 0). \tag{34}$$

The remainder is uniform over points whose neighborhoods have the same derivative bounds.

Proof. For fixed x, put $f_x(s) = \exp(-i\int_x^{x+s} A(r)\,dr)\psi(x+s)$. Then the two numerator terms in (33) are $f_x(h)$ and $f_x(-h)$. The derivative at zero is $f_x'(0) = \psi'(x) - iA(x)\psi(x)$. Taylor expansion of f_x at zero to second order with bounded third derivative gives $f_x(h) - f_x(-h) = 2hf_x'(0) + O(h^3)$. Division by $2h$ proves the assertion. The derivative bounds for A and ψ give the required uniform bound on f_x'''. □

A midpoint link $\widetilde{U}_h(x) = e^{-ihA(x+h/2)}$ has the same conclusion under these bounds. The midpoint quadrature error in the integral is $O(h^3)$, and $|e^{ia} - e^{ib}| \le |a-b|$ for real a, b transfers that error to the link. The two numerator perturbations are each $O(h^3)$ for bounded ψ, hence the stencil perturbation is $O(h^2)$.

Example 6.2 (Left-endpoint sampling can lose an order)**.** Take $A(x) = x$ near the origin and $\psi(x) = 1$. With the left-endpoint prescription $\widehat{U}_h(x) = e^{-ihA(x)}$, the stencil at zero is

$$\frac{\widehat{U}_h(0) - \widehat{U}_h(-h)^{-1}}{2h} = \frac{1 - e^{-ih^2}}{2h} = \frac{i}{2}h + O(h^3).$$

The continuum value $(\partial_x - iA(0))\psi(0)$ is zero. Thus this perfectly smooth input has a first-order error. The formula $U_j = e^{-ihA_j}$ by itself does not identify a second-order sampling scheme.

This qualification does not alter the finite Hamiltonian or its conservation laws. It isolates what is needed to interpret its link values as samples of a continuum connection. Nor does consistency of one stencil prove convergence of the coupled time evolution: stability, compatible initial data, regularity over the comparison interval, and a choice of reconstruction still have to enter that argument.

7 Non-Abelian currents and their frame dependence

Now take $\psi_x \in \mathbb{C}^2$, $U_{x,\mu} \in SU(2)$, and local gauges $g_x \in SU(2)$, as in the legacy non-Abelian chapter. The linked scalar

$$z_{x,\mu} = \psi_x^* U_{x,\mu}\psi_{x+\widehat{\mu}} \tag{35}$$

is gauge invariant: the adjoint factor g_x^* cancels g_x, and the factors at $x + \hat{\mu}$ also cancel. It follows that the source's scalar link observable

$$j_{x,\mu} = 2 \operatorname{Im} z_{x,\mu} \tag{36}$$

is invariant. To obtain a continuity equation with its particular sign and normalization one must specify the hopping generator. Invariance of the observable alone is not an equation of motion for color charge.

Let $\sigma^1, \sigma^2, \sigma^3$ be the Pauli matrices and define the real components $q_x^a = \psi_x^* \sigma^a \psi_x$. Expand the conjugated Pauli basis as

$$g_x^* \sigma^a g_x = \sum_b R_x{}^a{}_b \sigma^b, \qquad q_x^{g,a} = \sum_b R_x{}^a{}_b q_x^b. \tag{37}$$

The coefficients are real because conjugation preserves traceless Hermitian matrices. The trace inner product $\frac{1}{2}\operatorname{Tr}(AB)$ is preserved by conjugation, and the Pauli basis is orthonormal for it; hence $R_x R_x^{\mathsf{T}} = I$. Thus the color vector rotates while its Euclidean squared length stays fixed. The scalar charge $\psi_x^* \psi_x$ is separately invariant. A color component relative to a selected Pauli axis is not generally invariant under all local frames.

The same distinction applies to a plaquette: W changes by conjugation, while its trace does not. Keeping the full matrix retains transport information that a single trace can lose. Even for $SU(2)$, identical traces at separate plaquettes do not determine the relative frames needed to combine them into a larger loop.



Figure 4. Noncommuting $SU(2)$ factors retain their order. A based holonomy changes by conjugation, while the normalized Wilson trace is unchanged. The color fiber here is separate from the four-component Clifford spinor.

8 Exact ordered refinement of a two-by-two patch

The Abelian rule "multiply the fine plaquettes" hides two choices in a noncommutative group: where each plaquette is based, and in which order the transported matrices are multiplied. A 2×2 patch makes both choices visible without an asymptotic expansion.

Write X_{ij} for the backward horizontal link from $(i+1, j)$ to (i, j), where $i = 0, 1$ and $j = 0, 1, 2$. Write Y_{ij} for the backward vertical link from $(i, j+1)$ to (i, j), where $i = 0, 1, 2$ and $j = 0, 1$. Define the four fine plaquettes by

$$W_{ij} = X_{ij} Y_{i+1,j} X_{i,j+1}^{-1} Y_{ij}^{-1}, \qquad i, j \in \{0, 1\}. \tag{38}$$

Each W_{ij} acts at its own lower-left basepoint (i, j). The two row holonomies, based at $(0, j)$, are

$$R_j = (X_{0j} W_{1j} X_{0j}^{-1}) W_{0j}, \qquad j = 0, 1. \tag{39}$$

The conjugation transports the right plaquette into the left base fiber before multiplying it with the left plaquette.

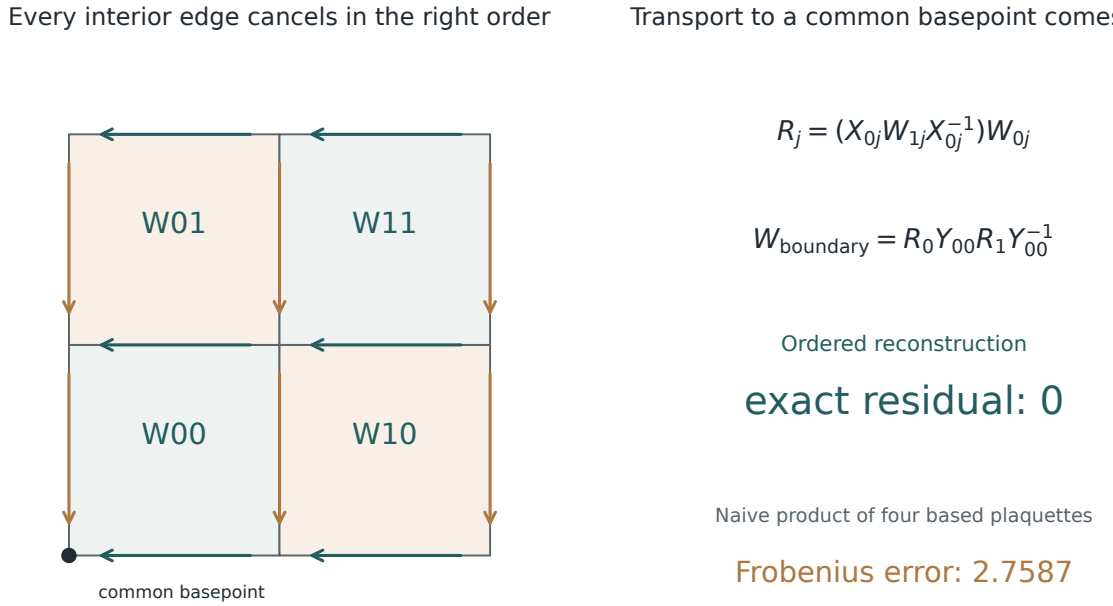


Figure 5. The four fine plaquettes are brought to common basepoints in a specified order. The exact cancellation leaves the eight outer links; a naive product omits the transport information.

Theorem 8.1 (Ordered patch identity)**.** *For invertible links in any matrix group, the coarse boundary holonomy satisfies*

$$W^{(2)} := X_{00} X_{10} Y_{20} Y_{21} X_{12}^{-1} X_{02}^{-1} Y_{01}^{-1} Y_{00}^{-1} \tag{40}$$

$$= R_0 Y_{00} R_1 Y_{00}^{-1}. \tag{41}$$

This identity requires neither small links nor commuting links.

Proof. First expand one row while retaining the product order:

$$\begin{aligned} R_j &= X_{0j}X_{1j}Y_{2j}X_{1,j+1}^{-1}Y_{1j}^{-1}X_{0j}^{-1}X_{0j}Y_{1j}X_{0,j+1}^{-1}Y_{0j}^{-1} \\ &= X_{0j}X_{1j}Y_{2j}X_{1,j+1}^{-1}X_{0,j+1}^{-1}Y_{0j}^{-1}. \end{aligned}$$

The adjacent pairs $X_{0j}^{-1}X_{0j}$ and $Y_{1j}^{-1}Y_{1j}$ cancel. Substituting these row expressions gives

$$\begin{aligned} R_0Y_{00}R_1Y_{00}^{-1} = X_{00}X_{10}Y_{20}X_{11}^{-1}X_{01}^{-1}\underbrace{Y_{00}^{-1}Y_{00}}_{I} \\ \cdot X_{01}X_{11}Y_{21}X_{12}^{-1}X_{02}^{-1}Y_{01}^{-1}Y_{00}^{-1}. \end{aligned}$$

Now $X_{01}^{-1}X_{01}$ cancels, followed by $X_{11}^{-1}X_{11}$. The remaining factors are exactly (40). □

All terms in (41) transform by conjugation at $(0,0)$ once their displayed transports are included. This checks the basepoint typing as well as the algebra. In the Abelian case conjugations disappear and the fine plaquettes can be reordered, recovering the familiar surface product. In the non-Abelian case a naive product of the four untransported matrices can be formed after identifying their coordinate spaces, but it has no reason to obey the required local gauge transformation or to equal the boundary holonomy.

This refinement is exact because it refers to one fixed set of fine links and its outer boundary. It does not assert that independently assigned coarse links agree with products of fine links. That separate compatibility question is the starting point of the geometric refinement estimates in the next part of the monograph.

9 Extracting a local curvature and cancelling its first error

Exact group multiplication and approximate Lie-algebra curvature are different operations. For fixed $A, B \in \mathfrak{su}(2)$ define the commutator loop

$$W_h = e^{hA}e^{hB}e^{-hA}e^{-hB}. \tag{42}$$

Here A and B are constant skew-Hermitian traceless matrices, and h is a real scalar near zero. Because $W_0 = I$, a single analytic matrix logarithm near the identity can be used for all sufficiently small $|h|$. Define

$$F_h = h^{-2}\log W_h, \qquad h \neq 0. \tag{43}$$

This construction describes a local constant-matrix model of noncommuting transport. A varying connection has additional derivative terms and must be analyzed with those terms present.

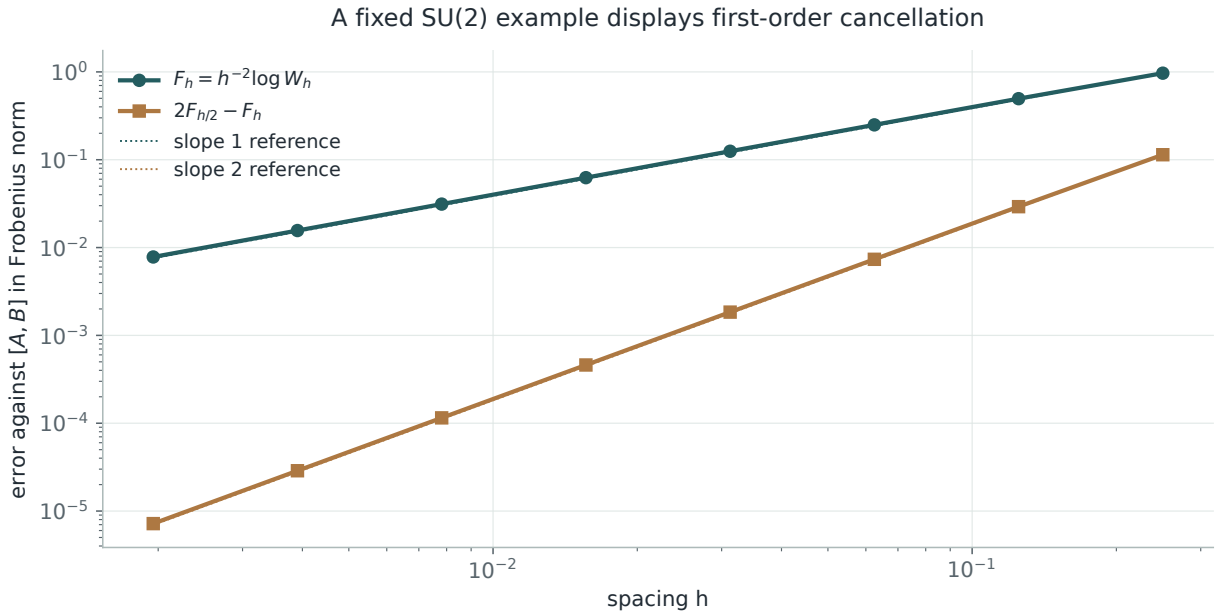


Figure 6. For the specified constant generators, the finite calculation illustrates first-order raw curvature error and cancellation of its leading term. The analytic statement is Proposition 9.1; a fitted slope over finitely many scales is a diagnostic of this example.

Proposition 9.1 (Local logarithmic refinement)**.** *On the common analytic logarithm branch, in any fixed matrix norm,*

$$F_h = [A, B] + hC_3(A, B) + O(h^2), \tag{44}$$

$$C_3(A, B) = \tfrac{1}{2}[A + B, [A, B]], \tag{45}$$

$$F_h^{[1]} := 2F_{h/2} - F_h = [A, B] + O(h^2). \tag{46}$$

For constant $G \in SU(2)$, simultaneous conjugation of A, B conjugates both F_h and $F_h^{[1]}$ by G.

Proof. Multiplying the four exponential power series through degree three gives

$$W_h = I + h^2[A, B] + \frac{h^3}{2}[A + B, [A, B]] + O(h^4).$$

The analytic logarithm has $\log(I + Z) = Z + O(\|Z\|^2)$ near zero. Since $Z = W_h - I = O(h^2)$, its correction begins at order h^4. Dividing by h^2 proves (44). At half spacing the linear term is $hC_3/2$, so $2F_{h/2} - F_h$ removes it exactly and leaves an $O(h^2)$ remainder. Finally, the power series for both the exponential and the local logarithm commute with conjugation. Applying them term by term proves the covariance statements. □

The common logarithm branch is essential: unrelated choices can introduce terms of size h^{-2} after rescaling and invalidate the cancellation. The Richardson value is a linear combination of Lie-algebra estimates; it is not asserted to be the logarithm obtained by the exact patch rule.

Geometry, defects, and reconstruction

10 Geometric half-edges and exact directional refinement

An exact change of resolution needs both a geometry and a comparison map. Replicating each coarse state into two spectator copies gives a useful algebraic test, but it does not make the translation visit a new point between neighboring sites. The geometric construction instead inserts an actual midpoint and transports the state to it.

We use the forward convention of the formal refinement modules. A link $u(x)$ transports from x to τx, so

$$u^g(x) = g(\tau x)u(x)g(x)^{-1}.$$

Thus $u(x) = U(x)^{-1}$ when $U(x)$ denotes the backward link used in the covariant difference stencil. Products act from right to left. All links in this part belong to $U(1)$; the unit modulus assumption is used in every Hilbert norm statement. Let $A = A^*$, $A^2 = 1$, act on $\mathbb{C}^4$, and put $P_\pm = (1 \pm A)/2$. The directional transport is

$$(S_{A,\tau,u}\psi)(x) = u(\tau^{-1}x)P_+\psi(\tau^{-1}x) + u(x)^{-1}P_-\psi(\tau x). \tag{47}$$

The channels are orthogonal. A bijective site translation and unit links therefore make this an isometry for the finite counting-space inner product established earlier.

10.1 Midpoint geometry and independent gauges

Write e_x for the retained endpoint and m_x for the midpoint of the edge $x \to \tau x$. The fine translation is the bijection

$$\sigma e_x = m_x, \qquad \sigma m_x = e_{\tau x}.$$

Consequently $\sigma^2 e_x = e_{\tau x}$ and $\sigma^2 m_x = m_{\tau x}$. This is the precise sense in which two half translations produce one coarse translation. Assigning half the edge spacing to σ is compatible with this combinatorics; it is not by itself a chart or continuum limit theorem.

Choose unit links $a(x)$ and $b(x)$ on the first and second half-edge with

$$b(x)a(x) = u(x). \tag{48}$$

The associated fine link is $v(e_x) = a(x)$, $v(m_x) = b(x)$. The interpolation and reconstruction maps are

$$(I_a\psi)(e_x) = \psi(x), \qquad (I_a\psi)(m_x) = a(x)\psi(x), \tag{49}$$

$$(R_a\Psi)(x) = \tfrac{1}{2}(\Psi(e_x) + a(x)^{-1}\Psi(m_x)). \tag{50}$$

The midpoint is a transported value, not an independently sampled field value. This identifies exactly the subspace on which the refinement theorem acts.

Coarse endpoint gauges $g(x)$ and midpoint gauges $q(x)$ may be chosen independently. The two sublinks transform by

$$a^g(x) = q(x)a(x)g(x)^{-1}, \qquad b^g(x) = g(\tau x)b(x)q(x)^{-1}.$$

The midpoint factors cancel in $b^g a^g$, leaving the required coarse transformation. At the midpoint,

$$a^g(x)g(x)\psi(x) = q(x)a(x)\psi(x),$$

while endpoint covariance is immediate. Hence interpolation commutes exactly with the corresponding gauge actions. This argument allows independent new gauge data at every inserted midpoint.

Theorem 10.1 (Exact two-half-step refinement)**.** *For every coarse state ψ and subdivision satisfying* (48)*,*

$$S^2_{A,\sigma,v} I_a \psi = I_a S_{A,\tau,u} \psi. \tag{51}$$

No flatness hypothesis is needed.

Proof. When the fine shift is applied twice, the mixed projector terms vanish because $P_+P_- = P_-P_+ = 0$. The remaining positive channel travels two half-edges forward, and the negative channel travels the same pair backward. At e_x, the positive coefficient is $b(\tau^{-1}x)a(\tau^{-1}x) = u(\tau^{-1}x)$; the negative coefficient is $a(x)^{-1}b(x)^{-1} = u(x)^{-1}$. The result is exactly (47). At m_x, the two surviving terms are

$$a(x)u(\tau^{-1}x)P_+\psi(\tau^{-1}x) + b(x)^{-1}P_-\psi(\tau x).$$

Since $b(x)^{-1} = a(x)u(x)^{-1}$, this is $a(x)$ times the coarse shifted state, as required by (49). □

10.2 The measure is part of the comparison

For finite sites, give each fine point half the coarse weight:

$$\langle \Psi, \Phi \rangle_{1/2} = \tfrac{1}{2} \sum_x \big(\langle \Psi(e_x), \Phi(e_x) \rangle + \langle \Psi(m_x), \Phi(m_x) \rangle \big).$$

Unitarity of $a(x)$ then gives

$$\langle I_a\psi, I_a\phi \rangle_{1/2} = \langle \psi, \phi \rangle\,, \qquad \langle I_a\psi, \Psi \rangle_{1/2} = \langle \psi, R_a\Psi \rangle\,. \tag{52}$$

The second identity follows by moving $a(x)$ from the first argument of the inner product to the second as $a(x)^{-1}$. It explains the averaging formula, including its factor $1/2$. In particular $R_aI_a = 1$ and I_aR_a is the orthogonal projection onto the transported midpoint subspace. Endpoint evaluation also gives a left inverse, but does not have this adjoint interpretation.

A local mass step can be refined factor by factor. If a local linear map N satisfies $N^2 = M$, then N commutes with the scalar interpolation. Consequently

$$N^2 S^2_{A,\sigma,v} N^2 I_a = I_a M S_{A,\tau,u} M.$$

Unitary spectral phases provide explicit choices of N. The refined expression replaces each coarse factor by its two half factors. It does not assert that two repetitions of a complete fine $NS_{A,\sigma,v}N$ walk equal the coarse walk: the intervening order would be different.

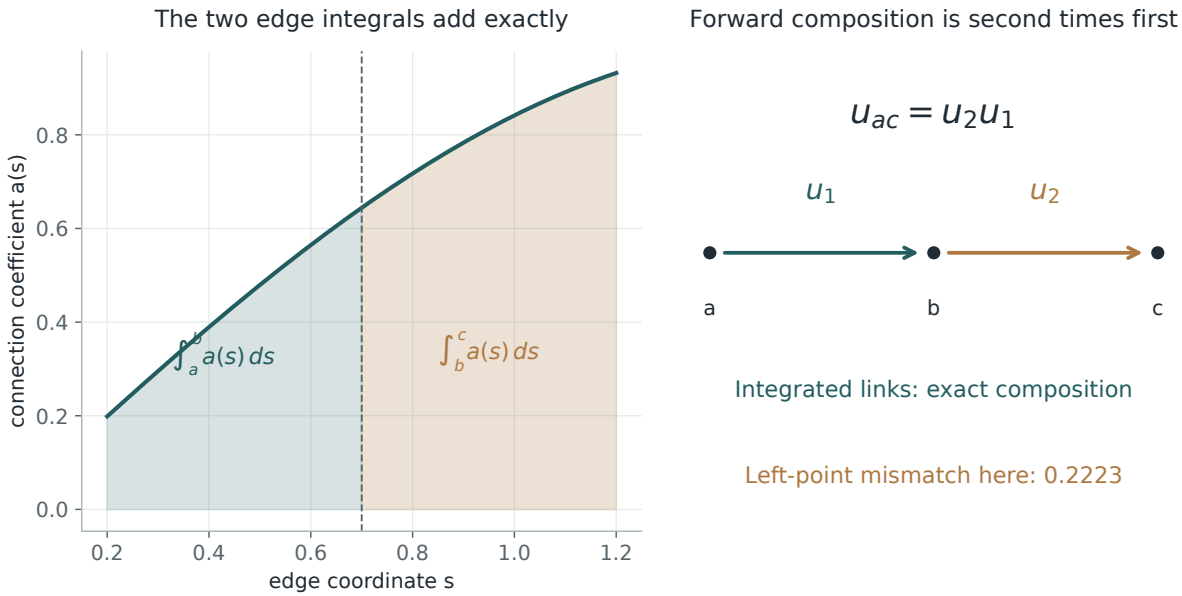


Figure 7. Integrated abelian phases compose exactly under edge subdivision. For the displayed coefficient $a(s) = \sin s$, the two subinterval integrals sum to the whole-edge integral. Independently sampled coarse and fine phases can disagree; that approximation error is separate from exact transport composition.

11 Cubical transport and the curvature obstruction

Three one-axis subdivisions do not automatically define a path-independent value at a shared corner. The issue already appears in one $2\times2\times2$ vertex cube. Let the vertices be $(x, y, z) \in \{0, 1\}^3$, with positive links $\ell_x(y, z)$, $\ell_y(x, z)$ and $\ell_z(x, y)$. There are twelve links and six faces. Under a vertex gauge, every link acquires its target gauge on the left and its source inverse on the right.

For example, the two positive paths across an xy face at height z are

$$p_{xy}(z) = \ell_y(1, z)\ell_x(0, z), \qquad p_{yx}(z) = \ell_x(1, z)\ell_y(0, z).$$

Define $\Omega_{xy}(z) = p_{xy}(z)p_{yx}(z)^{-1}$, and define the xz and yz face phases analogously. Both paths have the same endpoints, so their gauge factors cancel in this abelian ratio. Thus each Ω is gauge invariant, and $\Omega = 1$ precisely when the two path phases agree.

11.1 Six paths and one Bianchi relation

The three coordinate directions can be traversed in six orders, denoted $p_{XYZ}, p_{XZY}, p_{YXZ}, p_{YZX}, p_{ZXY}$ and p_{ZYX}. These names record execution order, while link multiplication still acts from right to left. Adjacent transpositions give

$$\begin{aligned} p_{XYZ} &= \Omega_{xy}(0)p_{YXZ}, & p_{XYZ} &= \Omega_{yz}(1)p_{XZY}, \\ p_{YXZ} &= \Omega_{xz}(1)p_{YZX}, & p_{XZY} &= \Omega_{xz}(0)p_{ZXY}, \\ p_{YZX} &= \Omega_{yz}(0)p_{ZYX}, & p_{ZXY} &= \Omega_{xy}(1)p_{ZYX}. \end{aligned}$$

Each identity follows by replacing the two-edge portion being swapped by its alternative path and its face ratio. The scalar nature of the links permits the common third edge to commute past the ratio.

Proposition 11.1 (Abelian cubical Bianchi identity)**.** *The six face phases satisfy*

$$\Omega_{yz}(1)\Omega_{xz}(0)\Omega_{xy}(1) = \Omega_{xy}(0)\Omega_{xz}(1)\Omega_{yz}(0). \tag{53}$$

If all six faces are flat, the six corner path phases agree. Flatness of the five faces other than $xy(1)$ already forces the sixth to be flat.

Proof. Use the three identities down the left route from XYZ through YXZ, YZX to ZYX, and the three down the right route through XZY, ZXY. They express p_{XYZ} as the respective products in (53) times p_{ZYX}. Cancel the unit terminal phase. If the five displayed independent face phases equal one, the identity gives $\Omega_{xy}(1) = 1$. When all faces are flat, every adjacent path swap leaves the phase unchanged, and the displayed relations connect all six paths. □

This is a local abelian identity. A matrix-valued version must account for the basepoint and transport of each face holonomy; the preceding formula cannot simply be copied with noncommuting matrices. Local flatness also says nothing by itself about holonomy around noncontractible loops in a larger lattice.

11.2 Interpolation is possible even when it depends on a path

Choose one reference path from the origin to each vertex and let $r(v)$ be its transport. Then

$$(I_\ell\psi)(v) = r(v)\psi, \qquad \langle \Psi, \Phi\rangle_{\text{cube}} = \tfrac{1}{8}\sum_{v\in\{0,1\}^3} \langle \Psi(v), \Phi(v)\rangle\,.$$

Every $r(v)$ is unitary, so this map is a Hilbert isometry from one spinor to the eight-child cell. Its gauge covariance follows from telescoping along the selected paths. Neither property requires flatness. What curvature obstructs is independence from the reference-path choice.

Suppose two such paths have phases p and q with $p = \Omega q$. Their discrepancy has the exact form

$$D(p, q; \psi) = p\psi - q\psi = q(\Omega - 1)\psi, \qquad \|D(p, q; \psi)\|^2 = |\Omega - 1|^2 \|\psi\|^2\,. \tag{54}$$

This is an equality, not a small-curvature approximation. It follows by factoring q and using $|q| = 1$. For a nonzero test spinor, the defect vanishes exactly when $\Omega = 1$. The formal nonflat cube has $p_{XYZ} = -1$ and $p_{YXZ} = 1$, producing a nonzero defect on the first basis spinor. Thus the distinction between covariant interpolation and path-independent interpolation is witnessed by explicit data.

For later estimates, retain the five-face quantity

$$\begin{aligned}\kappa(\ell) = {} & |\Omega_{xy}(0) - 1| + |\Omega_{xz}(1) - 1| + |\Omega_{yz}(0) - 1| \\ & + |\Omega_{yz}(1) - 1| + |\Omega_{xz}(0) - 1|.\end{aligned} \tag{55}$$

The omitted face is $xy(1)$. This selection is the one used by the formal five-swap composition; the multiplicities in later constants come from this choice.

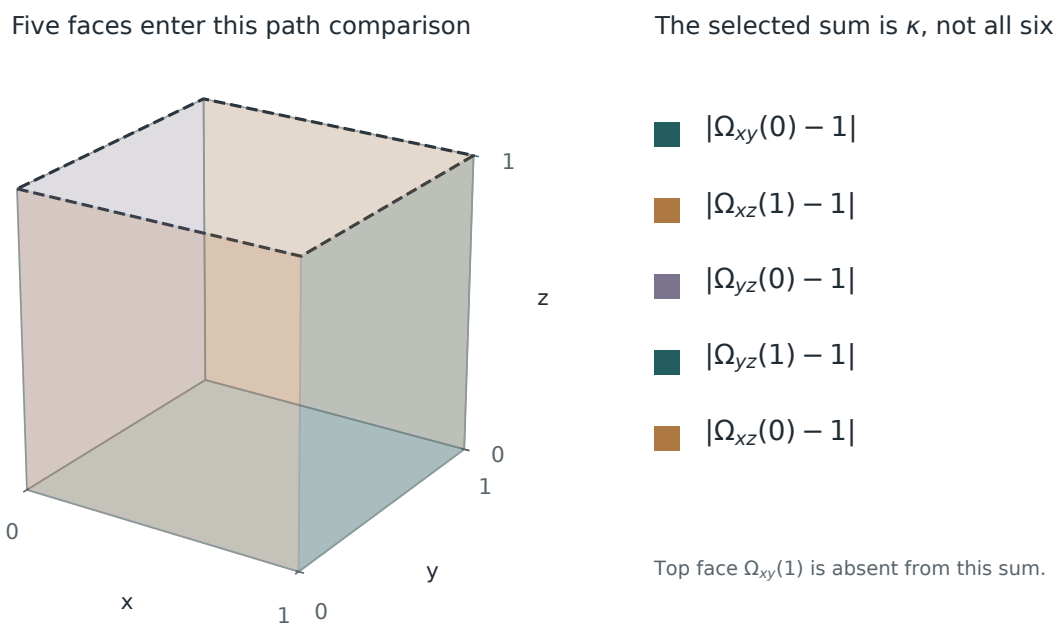


Figure 8. The selected five faces correspond to five adjacent swaps among the six corner paths. The sixth face is linked to them by the abelian cube Bianchi identity; it is not an independently omitted physical boundary condition.

12 Exact straight refinement and curvature are independent

The one-edge identity can be assembled into a three-axis refinement on the fine site set $\mathrm{Site} \times \{0,1\}^3$. The half translation σ_i toggles the corresponding child bit; when that bit carries from one to zero, it applies the coarse translation τ_i to the site. Thus σ_i^2 is the lift of τ_i. A phase frame $r(s,c) \in U(1)$ defines

$$(I_r\psi)(s,c) = r(s,c)\psi(s).$$

With one-eighth fine weights this is again an isometry.

12.1 The straight compatibility contract

For each direction, canonical fine links can be defined by

$$v_i(p) = r(\sigma_i p)\,\gamma_i(p)\,r(p)^{-1},$$

where $\gamma_i(p) = 1$ on a noncarrying half step and $\gamma_i(p) = u_i(s)$ on a carrying half step. Cancellation of the intermediate frame gives

$$v_i(\sigma_i p)v_i(p) = r(\sigma_i^2 p)u_i(s)r(p)^{-1}. \tag{56}$$

For arbitrary fine links, equation (56) is the explicit straight compatibility hypothesis. It is gauge covariant, and it is exactly the hypothesis needed to repeat the projector argument of (51):

$$S_{i,\mathrm{fine}}^2 I_r = I_r S_{i,\mathrm{coarse}}.$$

If the mass substeps also intertwine, the identities compose through the entire execution order $M - X - Y - Z - Y - X - M$. The refined walk here replaces each coarse spatial factor by two half factors and each coarse local factor by its supplied subdivision. The result is exact on the framed subspace.

The construction's internal cube links are pure frame ratios and therefore locally flat. Nevertheless, when the coarse x and y translations commute, the large two-by-two fine Wilson loop equals the coarse plaquette. Coarse curvature survives the refinement. Hence exact straight refinement can preserve a nontrivial coarse plaquette; it does not require that every loop become flat.

12.2 A flat cube need not reconstruct the prescribed edge

Conversely, take an arbitrary ordered half-link pair (a, b) and call its actual product $v = ba$. It is always a subdivision of v, even when the separately prescribed coarse link is $u \neq v$. The exact directional defect is

$$\begin{aligned} D_{\text{fine}}\psi &:= S^2_{A,\sigma,(a,b)} I_a \psi - I_a S_{A,\tau,u}\psi \\ &= I_a (S_{A,\tau,v} - S_{A,\tau,u})\psi. \end{aligned} \tag{57}$$

The equality follows by applying the exact theorem to the product v and using linearity of I_a. Endpoint restriction makes I_a injective, so the fine defect vanishes on a state precisely when its coarse mismatch vanishes on that state. This statewise statement is not a claim that one test state detects every link error.

A concrete counterexample removes any possible inference from flatness alone. Let both half-links be i, and prescribe $u = 1$. Their product is $v = -1$. A cube whose twelve positive links are all i has every face plaquette equal to one. On a one-site periodic lattice, however, $\tau = 1$ and

$$S_{A,1,1} = P_+ + P_- = 1, \qquad S_{A,1,-1} = -P_+ - P_- = -1.$$

Thus the coarse mismatch is -2ψ, and the normalized fine defect has norm $2\,\|\psi\|$. The same link data are perfectly flat in the path-comparison sense and maximally wrong relative to the prescribed straight link. This is the role of the quarter-turn witness in the formal source.

Two questions must therefore remain separate: do the ordered fine links reconstruct the desired coarse edge, and do alternative routes between the same endpoints agree? The first is an adjacent-scale compatibility question; the second measures curvature. A reliable error estimate keeps both contributions visible.

13 Hilbert estimates with explicit constants

Finite-dimensional continuity is not enough to identify an operator norm with physical ℓ^2 energy. The earlier continuous-linear packaging retains the ambient norm on a function space. The Hilbert scalar layer instead realizes a finite four-spinor field as

$$\mathcal{H}_S = \ell^2(S \times \{0,1,2,3\}; \mathbb{C}), \qquad \|\psi\|^2 = \sum_{x \in S} \|\psi(x)\|^2_{\mathbb{C}^4}.$$

The formal realization is `EuclideanSpace`; its norm is identified explicitly with the real part of the previously defined state inner product. All operator norms from this point use this Hilbert structure. Assume S is finite and nonempty when taking a maximum over sites.

Theorem 13.1 (Constant-one directional mismatch bound). *Let A be a Hermitian Clifford involution, τ a site bijection, and $u, v : S \to U(1)$. Then*

$$\left\|S_{A,\tau,v} - S_{A,\tau,u}\right\| \le \delta(u,v) := \max_{x\in S} |v(x) - u(x)| = \max_{x\in S} |v(x)u(x)^{-1} - 1|. \tag{58}$$

The constant is independent of the number of sites. The actual half-grid defect in (57), with half-cell Hilbert measure, obeys the same bound.

Proof. At each output site the difference of the shifts has a positive-channel term and a negative-channel term. Orthogonality gives the exact energy identity

$$\begin{aligned}\|(S_v - S_u)\psi\|^2 = \sum_x [&|v(\tau^{-1}x) - u(\tau^{-1}x)|^2 \left\|P_+\psi(\tau^{-1}x)\right\|^2 \\ &+ |v(x)^{-1} - u(x)^{-1}|^2 \left\|P_-\psi(\tau x)\right\|^2].\end{aligned}$$

There is no mixed term, even though the two spinors come from different sites: they lie in orthogonal fixed channel ranges. Unit modulus implies $|v^{-1} - u^{-1}| = |v - u|$. Bound every coefficient by δ^2, reindex the two sums through τ, and use $\|P_+\eta\|^2 + \|P_-\eta\|^2 = \|\eta\|^2$. The result is at most $\delta^2 \|\psi\|^2$. Taking square roots and then the operator supremum proves the first inequality. Multiplication by $u(x)^{-1}$ proves the equality of scalar maxima. Finally (57) and the isometry of I_a identify the fine defect norm on every input with the coarse mismatch norm. □

In ordinary Euclidean coordinates on the fine site set, the half-cell measure is implemented by multiplying the interpolated coordinates by $1/\sqrt{2}$. Omitting this factor changes the norm and inserts an artificial $\sqrt{2}$ into the comparison. The quarter-turn example attains the value two, showing that the constant one cannot be reduced uniformly over all allowed data.

13.1 The telescoping mechanism for a complete walk

Let $I : X \to Y$ be a bounded linear comparison map and let $F_j : Y \to Y$, $C_j : X \to X$ be fine and coarse factors. Write $D_j = F_j I - I C_j$. Two factors satisfy the exact identity

$$F_2 F_1 I - I C_2 C_1 = F_2 D_1 + D_2 C_1.$$

Repeated insertion gives, in execution order $1, \dots, N$,

$$F_N \cdots F_1 I - I C_N \cdots C_1 = \sum_{j=1}^{N} F_N \cdots F_{j+1} D_j C_{j-1} \cdots C_1. \tag{59}$$

Empty products are identities. If every F_j, C_j is contractive, the triangle inequality and submultiplicativity yield

$$\|F_N \cdots F_1 I - I C_N \cdots C_1\| \le \sum_{j=1}^{N} \|D_j\|.$$

No separate bound on I is needed for this form, since its contribution is already present inside each D_j. Exact local intertwining makes every term zero.

For the scalar seven-factor estimate, both walks act on the same finite Hilbert space, with $I = 1$. One uses the reconstructed links v_x, v_y, v_z, the other the prescribed links u_x, u_y, u_z, and both use the same

two contractive mass factors M_0, M_1. With execution order $M_0 - X - Y - Z - Y - X - M_1$, define $\delta_i = \delta(u_i, v_i)$. The mass defects vanish. Equation (59) and (58) prove

$$\|W_v - W_u\| \le 2\delta_x + 2\delta_y + \delta_z. \tag{60}$$

The repeated X and Y factors explain the coefficients two. The upper bound need not be attained by the complete walk: its individual error terms can cancel. It is an operator estimate between effective common-space walks. The separate half-grid theorem supplies the genuine normalized fine directional estimate; a general spatially varying three-axis comparison still has to provide its actual embeddings and local factor defects.

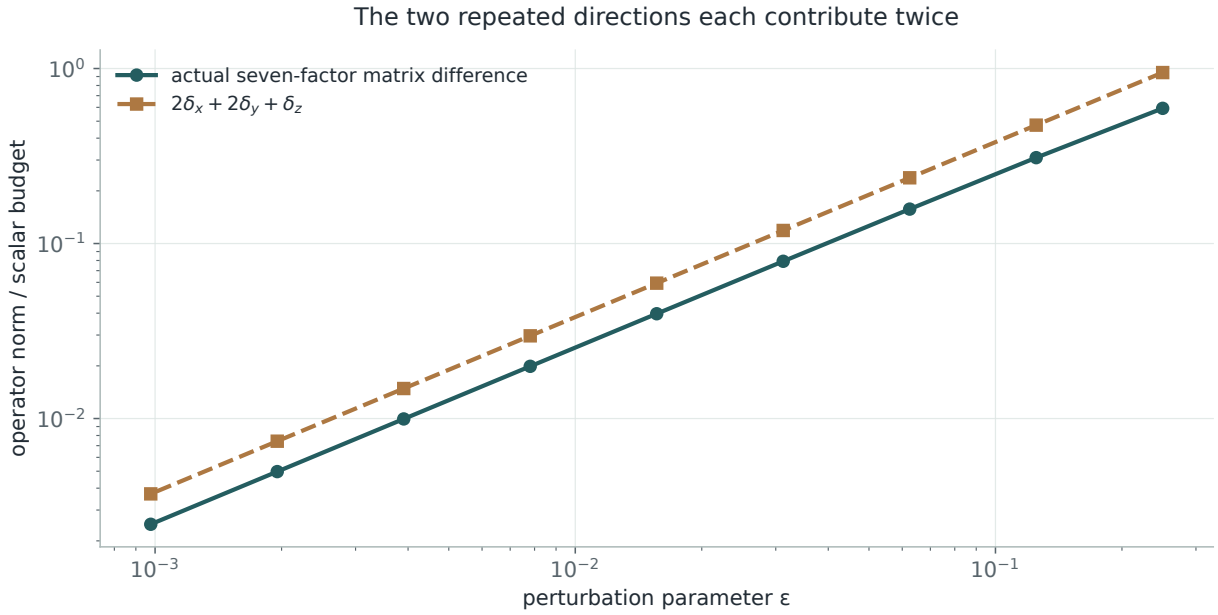


Figure 9. A finite matrix illustration of the telescoping mechanism and the repeated-factor budget. The plotted common-space example tests the algebraic estimate; it is not an additional formal lattice-to-continuum theorem.

14 A two-stage edge and path estimate

The cube phase discrepancies can be made into bounded operators on the same finite Hilbert state space. For $z \in U(1)$, let $P_z\psi = z\psi$ uniformly at every site. If $p = \Omega q$, then

$$P_p - P_q = q(\Omega - 1)\mathrm{id}, \qquad \|P_p - P_q\| = |\Omega - 1|.$$

The norm equality uses a nonzero Hilbert space. It is the operator form of (54).

The formal construction selects five ordered phase pairs, in this order:

$$(p_{XYZ}, p_{YXZ}), \quad (p_{YXZ}, p_{YZX}), \quad (p_{YZX}, p_{ZYX}), \quad (p_{XYZ}, p_{XZY}), \quad (p_{XZY}, p_{ZXY}).$$

Let Q_f be the execution-order product of the first phase operators in these pairs, and Q_c the product of the second ones. Identity factors can be placed at the ends to fit the seven-factor interface. Applying (59) yields

$$\|Q_f - Q_c\| \le \kappa(\ell).$$

These are explicitly constructed products of uniform phase operators. Calling them a path-reordering stage describes their algebraic role; it does not assert that a global spatially varying inter-cell assembly has already been constructed.

Theorem 14.1 (Combined finite Hilbert estimate)**.** *With the common-space walks and contractive mass factors from* (60)*, define*

$$D_{\text{total}} = Q_f W_v - Q_c W_u.$$

Then

$$\|D_{\text{total}}\| \le 2\delta_x + 2\delta_y + \delta_z + \kappa(\ell). \tag{61}$$

If $v_i = u_i$ for all three directions and the cube is flat, the defect is exactly zero.

Proof. Insert and subtract $Q_f W_u$:

$$D_{\text{total}} = Q_f(W_v - W_u) + (Q_f - Q_c)W_u.$$

All phase factors are unitary, and W_u is a product of contractions. Taking norms bounds the first term by the straight-edge budget and the second by the path budget. In the exact sector the walks agree and every selected phase pair agrees, so both terms vanish before taking norms. □

The sum keeps the logical independence proved in Section 12. A bound on face curvature cannot replace a bound on reconstruction ratios. Conversely, exact ordered products do not remove a nonflat alternative-path discrepancy. Neither conclusion requires that the estimate be sharp for every configuration.

15 Connection-generated links and a quadratic mesh bound

The scalar budgets become useful when links arise from a specified background rather than independent phases. Consider real abelian connection components A_x, A_y, A_z on a Euclidean chart. Define exact coordinate-edge angles and forward links by

$$\theta_i(p,h) = \int_0^h A_i(p + te_i)\,dt, \qquad u_i(p,h) = e^{i\theta_i(p,h)}. \tag{62}$$

The coupling is absorbed into A_i. These links are automatically unitary, and the interval integral is oriented when $h < 0$.

15.1 Exact composition precedes every estimate

Continuity of A_i along its own coordinate lines permits interval-integral additivity and a change of variable:

$$\begin{aligned}\theta_i(p,h) &= \int_0^{h/2} A_i(p+te_i)\,dt + \int_{h/2}^{h} A_i(p+te_i)\,dt \\ &= \theta_i(p,h/2) + \theta_i(p+(h/2)e_i, h/2).\end{aligned}$$

Exponentiation gives the ordered identity

$$u_i(p+(h/2)e_i, h/2)u_i(p,h/2) = u_i(p,h). \tag{63}$$

Thus the reconstructed and prescribed links agree exactly and all three straight-edge errors vanish. This is an additivity theorem for exact line integrals. Independently approximating each integral by a quadrature rule would introduce another error term, which is not silently included in the present bound.

For an affine component, $A_i(p+te_i) = A_i(p) + t\,\partial_i A_i$, so

$$\theta_i(p,h) = hA_i(p) + \tfrac{1}{2}h^2\partial_i A_i.$$

The face angle is exactly h^2F_{ij}, where $F_{ij} = \partial_i A_j - \partial_j A_i$. Hence $\Omega_{ij} = e^{ih^2F_{ij}}$. Using $|e^{it}-1| \le |t|$ and the selected five faces yields

$$\|D_{\text{total}}(h)\| \le h^2(|F_{xy}| + 2|F_{xz}| + 2|F_{yz}|). \tag{64}$$

This exact affine calculation is stronger than a truncated Taylor argument. It also fixes the orientation sign of the curvature convention.

15.2 The actual coordinate regularity assumptions

The more general formal connection structure records the three components, the six cross derivatives $\partial_i A_j$ for $i \neq j$, and their one-dimensional derivative identities along the relevant coordinate lines. Each component is continuous along its own integration direction. Each cross derivative is continuous along the line in its differentiation direction. Finally there are global nonnegative constants

$$|\partial_i A_j(p)| \le B_{j|i} \qquad (i \neq j).$$

For example, $B_{y|x}$ bounds the derivative of A_y in the x direction. These explicit hypotheses are sufficient for the coordinate arguments below. They avoid appealing to an unspecified stronger multivariable smoothness class.

The fundamental theorem of calculus first gives

$$|A_j(p+he_i) - A_j(p)| \le B_{j|i}|h|.$$

Integrating along the transverse j edge then gives

$$|\theta_j(p+he_i, h) - \theta_j(p,h)| \le B_{j|i}|h|^2. \tag{65}$$

Indeed the integrand difference is bounded by $B_{j|i}|h|$ at every point of an interval of length $|h|$. Oriented integration changes the sign but not this absolute bound.

For the xy face, the two-path angle difference is

$$\begin{aligned}\Theta_{xy}(p,h) = \theta_y(p+he_x,h) - \theta_y(p,h)\\ - (\theta_x(p+he_y,h) - \theta_x(p,h)).\end{aligned}$$

Therefore

$$|\Theta_{xy}(p,h)| \le (B_{y|x} + B_{x|y})h^2, \qquad |\Omega_{xy}(p,h) - 1| \le (B_{y|x} + B_{x|y})h^2.$$

For the last implication, $e^{it} - 1 = \int_0^t ie^{is}\,ds$ has absolute value at most $|t|$. The other two plane estimates follow from their corresponding cross derivatives. This proof bounds absolute derivatives separately; it does not prove a sharper general surface-integral formula that exploits cancellation inside F_{ij}.

Theorem 15.1 (Regular background estimate)**.** *Suppose the coordinate regularity and derivative bounds above hold. Set*

$$\begin{aligned}\mathcal{A}_\partial = B_{y|x} + B_{x|y} + 2(B_{z|x} + B_{x|z})\\ + 2(B_{z|y} + B_{y|z}).\end{aligned} \tag{66}$$

Generate the reconstructed and prescribed finite-site links from the exact integrals (62)*, and generate the twelve links of the comparison cube from the same connection at a chart point p. For the two-stage finite Hilbert operator of Section 14, with the same contractive mass factors on both sides,*

$$\|D_{\text{total}}(h)\| \le h^2\mathcal{A}_\partial. \tag{67}$$

Proof. Equation (63) makes $\delta_x = \delta_y = \delta_z = 0$. Apply the appropriate face estimate to each term in (55). There is one xy term, two xz terms and two yz terms, giving $\kappa \le h^2\mathcal{A}_\partial$. Substitute in (61). □

The formal theorem permits an arbitrary finite nonempty site type, three site bijections and a chart-position map, along with an independently specified cube basepoint. It does not impose a geometric identity between every site translation and that position map. The result is consequently an exact estimate for the declared generated operators; interpreting them as one global geometric lattice requires those additional compatibility choices.

15.3 A spatially varying witness

Take $A_x = A_z = 0$ and $A_y(x,y,z) = \sin x$. Its curvature $F_{xy} = \cos x$ is nonconstant. The only nonzero derivative envelope is $B_{y|x} = 1$, so $\mathcal{A}_\partial = 1$. Here

$$\theta_y(p,h) = h\sin x, \qquad \Omega_{xy}(p,h) = \exp\big(ih[\sin(x+h) - \sin x]\big).$$

The other plane holonomies are one, and the two-stage defect is bounded by h^2. The bound is uniform in the basepoint. The trigonometric formula illustrates how actual phases can be substantially smaller than a global envelope; it does not change the envelope's role in a uniform theorem.

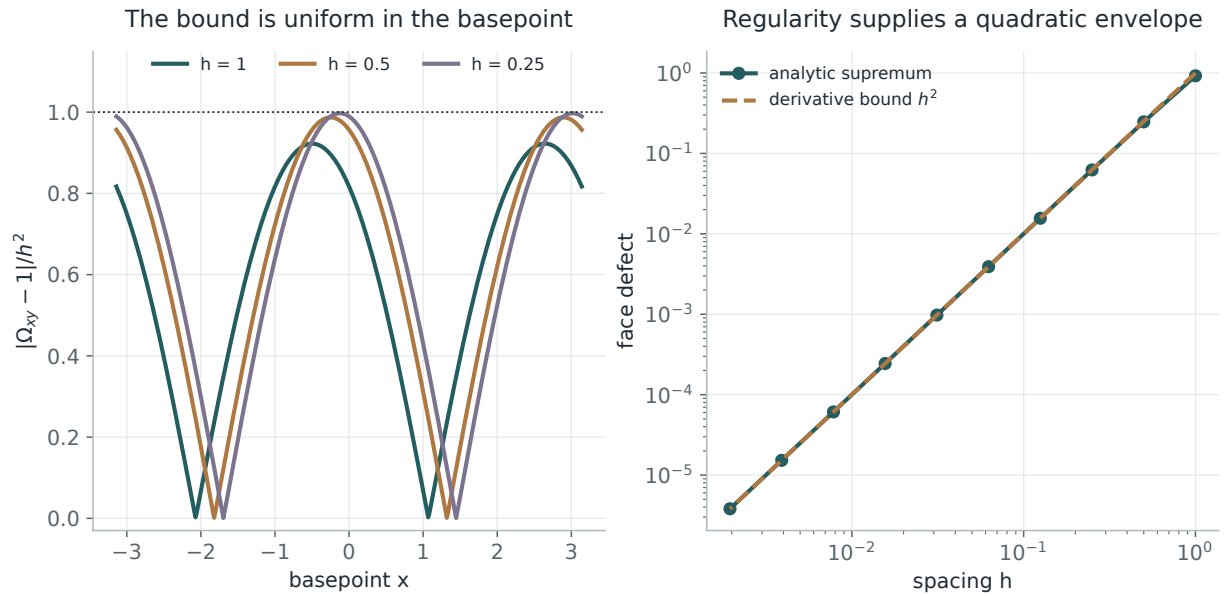


Figure 10. The non-affine background $A_y = \sin x$ has varying curvature and exact line-integral links. The quadratic envelope follows from $|\partial_x A_y| \le 1$; the computed phase amplitudes illustrate this bound over the plotted finite range.

16 Summable refinement and what a limit still requires

For the dyadic mesh $h_n = a_0 2^{-n}$, the regular estimate produces the nonnegative sequence

$$\varepsilon_n = C4^{-n}, \qquad C = a_0^2 \mathcal{A}_\partial.$$

The actual generated operator norms are bounded by this sequence and hence are summable. This is more information than merely $\varepsilon_n \to 0$. A sequence with adjacent errors tending to zero need not converge; summability controls the accumulated tail.

Proposition 16.1 (A complete-space limit with an explicit tail)**.** *Let f_n be a sequence in a complete metric space and suppose*

$$d(f_n, f_{n+1}) \le C4^{-n} \qquad (n \ge 0), \quad C \ge 0.$$

Then f_n converges to some f_∞, and

$$d(f_n, f_\infty) \le \sum_{k=n}^{\infty} C4^{-k} = \tfrac{4}{3} C4^{-n}. \tag{68}$$

Proof. For $m > n$, the triangle inequality gives

$$d(f_n, f_m) \le \sum_{k=n}^{m-1} C4^{-k} \le \tfrac{4}{3} C4^{-n}.$$

The upper bound tends to zero with n, so the sequence is Cauchy. Completeness provides a limit. Passing $m \to \infty$ in the distance estimate proves the tail bound, and the displayed evaluation is the geometric series. □

The formal regular-connection limit theorem has precisely this conditional shape: it takes a sequence in a named complete metric space and an adjacent-distance hypothesis bounded by the regular dyadic budget. The source also proves summability of the actual connection-generated finite-operator norms. These are two compatible ingredients, but the operator bound alone does not manufacture the sequence f_n or identify its terms across different Hilbert spaces.

For a physical refinement application one must define the reconstructed states or evolutions in a common space and derive the required distance estimate from the relevant operator defects. If the input state is not normalized, its norm enters that derivation. If the number of time steps grows with the refinement level, the accumulation in time must also be estimated. A limit supplied by this abstract argument is not automatically a Dirac propagator or a solution of a coupled gauge-field system.

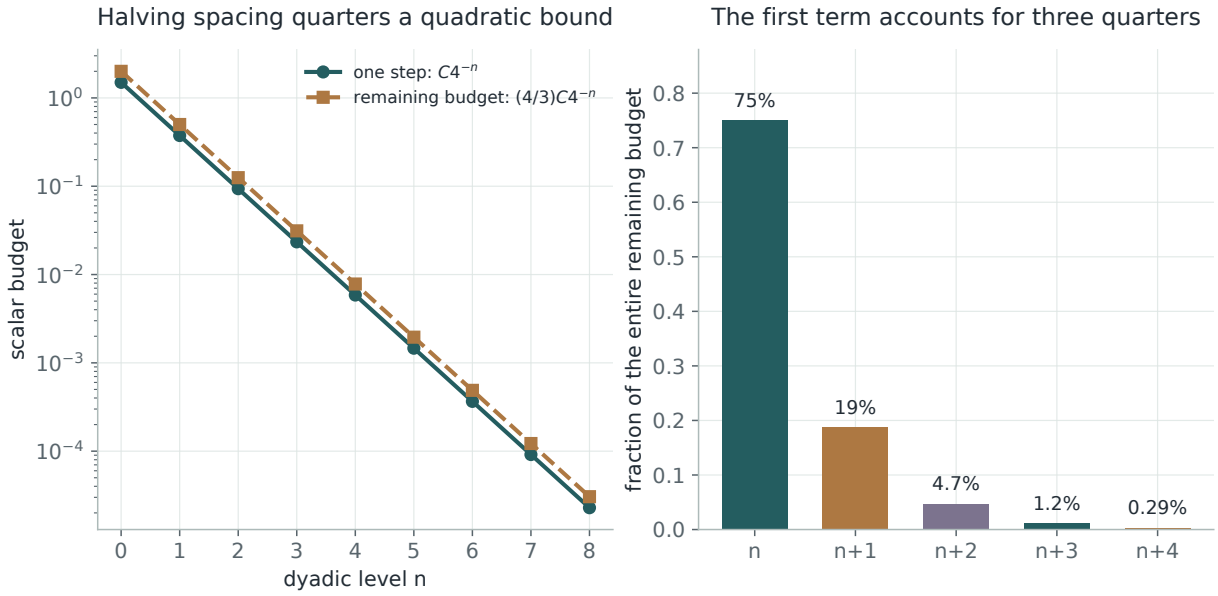


Figure 11. For the regular connection envelope, halving the mesh quarters the one-level budget. The remaining accumulated error is bounded by the geometric tail $\frac{4}{3}C4^{-n}$ once the adjacent-distance hypothesis has been established in a common complete space.

17 From weighted cells to a continuum Hilbert space

The reconstruction blocks that follow the connection estimates supply an important part of the common-space infrastructure. They establish what finite cell coefficients mean as L^2 fields, then identify a concrete family of approximating subspaces. They do not by themselves transfer the gauge-coupled walk or its generator through those maps.

17.1 Cell values and normalized coefficients

For a measurable finite partition into cells of equal volume $v > 0$, distinguish the physical constant value $q_a \in \mathbb{C}^4$ on cell C_a from the normalized coefficient $\psi_a = \sqrt{v}\, q_a$. The step interpolation is

$$(J\psi)(x) = v^{-1/2}\psi_a \quad (x \in C_a),$$

with equality understood almost everywhere. Direct integration gives

$$\|J\psi\|_{L^2}^2 = \sum_a \int_{C_a} v^{-1} \|\psi_a\|^2 = \sum_a \|\psi_a\|^2 .$$

Equivalently, physical cell values carry the weighted norm $v \sum_a \|q_a\|^2$. The adjoint averaging map yields the orthogonal projection onto the finite step subspace. Its Pythagorean decomposition proves best approximation in L^2.

At octree depth n, the number of cells is 8^n, their side scale is 2^{-n} and their volume is 8^{-n}. The first construction uses infinite octant addresses with the uniform product probability measure. Finite-prefix step subspaces are nested and have dense union. The weighted normalization is thus tied to the actual measure of a prefix cell, rather than inferred from the number of coordinates alone. The later source convention `euclideanDyadicLevel d` uses prefix depth $d+1$; the geometric depth n used here counts digits directly.

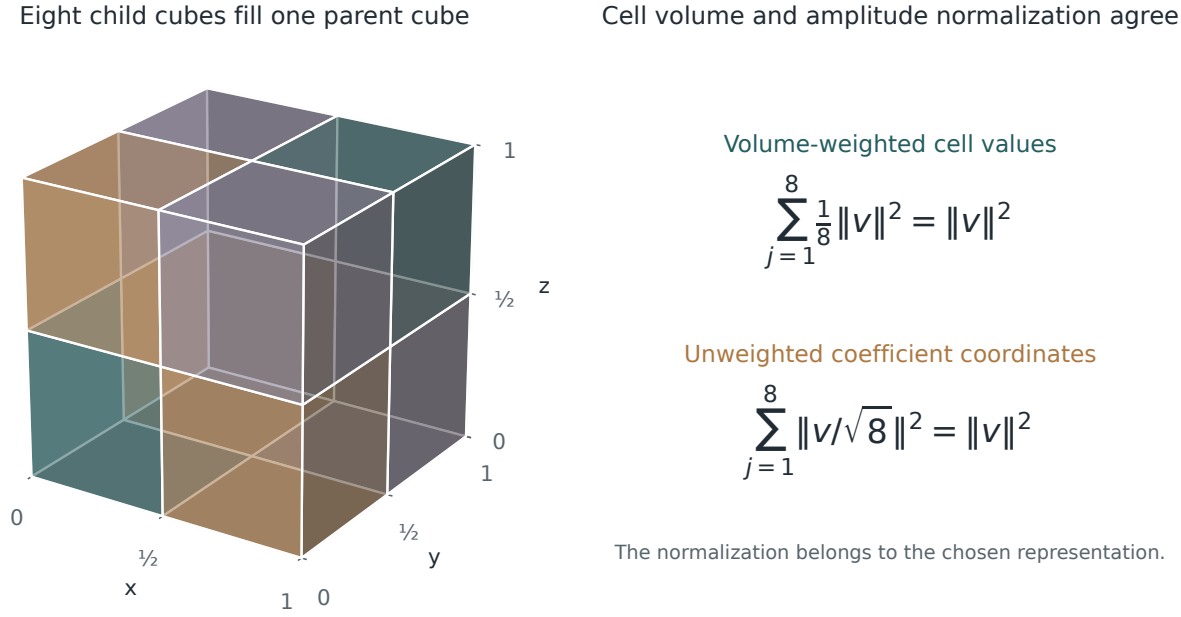


Figure 12. Eight equal children preserve a parent cell's total volume. Constant transported values use child volume weights; counting-normalized coefficients instead include the square root of the child volume. These are equivalent descriptions of the same Hilbert normalization.

17.2 The geometric identification is an additional theorem

Splitting each octant digit into three bits and applying positional binary decoding gives a continuous surjection from address space onto the unit cube. The decoder is not pointwise injective: dyadic coordinates admit the familiar terminating and nonterminating binary descriptions. The measure result then proves that fair binary product measure pushes forward to cube volume, producing a norm-preserving L^2 pullback.

At the intermediate stage, finite-resolution Euclidean subspaces are defined by adjoint transport of address-space levels. Density and strong convergence of their orthogonal projections can be proved at that stage, but it would be premature to call them ordinary dyadic-box subspaces. The next block closes

this identification: a prefix cell equals the preimage of its corresponding concrete Cartesian dyadic box almost everywhere. The formal boxes are closed and share boundary faces; those faces have zero volume. Equality in L^2 therefore survives the binary boundary ambiguity.

The transported and concrete closed levels are then proved equal. Their union is dense, and the orthogonal projections P_n satisfy

$$\|P_n f - f\|_{L^2} \longrightarrow 0 \qquad \text{for every } f \in L^2([0,1]^3; \mathbb{C}^4).$$

This is a state-space approximation theorem. Strong convergence of projections does not control an unbounded Dirac operator, establish graph convergence, or identify the limit of gauge-coupled propagators. Those statements need consistency on suitable domains and stability estimates for the transported operators.

The resulting position is precise. Exact covariant transport, genuine geometric subdivisions, finite Hilbert defect bounds, connection-derived mesh estimates and concrete continuum approximation spaces are available as separate, source-backed layers. Their assembly into a regular-background physical Hamiltonian remains a further mathematical task. The self-adjoint free-torus Hamiltonian provides an operator endpoint for the ungauged branch; importing these background estimates into that setting requires explicit domain and comparison-map work.

18 Gauge transport and its continuum comparison

Endpoint cancellation gives covariance of composed transport and conjugation of closed holonomy. In the specified finite Maxwell–Dirac system, matter continuity, Gauss propagation, and electric work fit together exactly. Ordered non-Abelian products retain the path information required by refinement.

Geometric interpolation supplies actual midpoints and weighted isometries. Straight-edge reconstruction and path curvature enter distinct error budgets for explicitly constructed finite operators. Regular line-integral links reconstruct straight edges exactly, while derivative envelopes give summable dyadic bounds for the chosen curvature defect.

Correctly normalized finite fields become L^2 step functions. Concrete dyadic boxes identify the approximating subspaces modulo null boundaries, and their projections converge strongly on each fixed field. Comparison of physical generators and evolution remains a separate task.

18.1 The remaining analytic bridge

The next background-coupled result needs a common operator setting. One must specify the physical connection, its regularity, the continuum spinor domain, the discrete operators, and the embeddings that compare them. Consistency on a controlled core, stability, and a suitable graph or resolvent argument then have to connect the finite refinement estimates to the continuum operator. Each limit must identify its topology and its dependence on the state or its regularity.

The free Dirac route already provides a useful endpoint for this comparison. The free construction gives a self-adjoint physical torus Hamiltonian on its weighted Fourier domain. The generator chapter supplies its spectral-evolution identification as a written proof. Its formal certification remains distinct from the recorded operator certificate. A background gauge extension must account for those operator-domain and evolution questions as well as the link estimates developed here.

18.2 From a prescribed connection to coupled dynamics

A background estimate controls a supplied field. A coupled gauge theory makes that field part of the unknown. The finite semidiscrete example illustrates the desired compatibility among matter transport, electric evolution, and a Gauss constraint. Extending this compatibility to a specified continuum setting requires an existence theory, constraint propagation, and a convergence statement for the coupled system. Local covariance alone supplies none of these additional conclusions.

The constrained-gravity chapter examines the additional obligations imposed by a dynamical geometry: regular-chart reduction, constraint closure, boundary terms, and strong-field evolution. Arithmetic refinement and cosmological transport retain the same requirement to identify their state spaces and comparison maps, without sharing all of the gauge model's assumptions.

> **The next theorem should name its comparison**
>
> Specify the two state spaces, the operators or evolutions to be compared, the map between resolutions, and the norm or topology in which the defect tends to zero. Then state which connection and domain hypotheses make that comparison valid.

Sources and further reading

Project sources

[P1] Komorebi P. *Projector Refinement Physics, Volume I*. The Abelian gauge and ordered non-Abelian holonomy studies supply the finite field equations and matrix identities. These are written mathematical arguments. Exact formal coverage is identified separately.

[P2] Komorebi P. *Komorebi Physics formal source*. The gauge verification appendix records the transport and refinement declarations, their hypotheses, and the limitations of the retained compiler evidence.

[P3] Komorebi P. *Projectors, Packets, and Transfer* and *Fourier Packets and the Continuum Dirac Operator*. The packet framework and free-field operator construction in this book. Extending the latter to a background connection requires the domain and evolution comparisons stated here.

Primary context

[C1] C. N. Yang and R. L. Mills. "Conservation of Isotopic Spin and Isotopic Gauge Invariance." *Physical Review* **96** (1954), 191–195. https://doi.org/10.1103/PhysRev.96.191.

[C2] Kenneth G. Wilson. "Confinement of quarks." *Physical Review D* **10** (1974), 2445. https://doi.org/10.1103/PhysRevD.10.2445.

[C3] John Kogut and Leonard Susskind. "Hamiltonian formulation of Wilson's lattice gauge theories." *Physical Review D* **11** (1975), 395. https://doi.org/10.1103/PhysRevD.11.395.

[C4] Pablo Arnault and Fabrice Debbasch. "Quantum walks and discrete gauge theories." *Physical Review A* **93** (2016), 052301. https://doi.org/10.1103/PhysRevA.93.052301.

[C5] C. Cedzich, T. Geib, A. H. Werner, and R. F. Werner. "Quantum walks in external gauge fields." *Journal of Mathematical Physics* **60** (2019). Author preprint arXiv:1808.10850v1, 31 August 2018. https://arxiv.org/abs/1808.10850v1; https://doi.org/10.1063/1.5054894.

How these sources are used

Yang and Mills provide the original continuum setting for local non-Abelian covariance. Wilson and Kogut–Susskind identify the lattice and Hamiltonian settings in which links, closed-loop observables, electric variables, and constraints belong. These references provide context; their quantization or confinement results are not premises of the elementary identities proved here. A gauge-invariant expression for a walk does not by itself supply a Hamiltonian phase space for its gauge field.

Arnault–Debbasch and Cedzich–Geib–Werner–Werner are prior work on gauge-covariant discrete-time walks. In the latter preprint, equations (19)–(20) give endpoint covariance and conjugation of closed transport, while Section IV.6 explains why coupling depends on the ordered decomposition of a walk. These are direct points of comparison with the conventions and factorwise refinement used here. Their plaquette in equation (22) traverses the opposite orientation to the ratio of forward paths used in this volume; in the Abelian case the two phases are inverses. Its global lattice statements should not be transferred to a periodic quotient without accounting for noncontractible loop holonomy.

The source-check record separates inspected full text from bibliographic authentication. Relevant full-text sections of the Yang–Mills article and the Cedzich et al. preprint were inspected. For Wilson, Kogut–Susskind, and Arnault–Debbasch, the accompanying record authenticated the publisher metadata and abstract, but full-text access through the publisher was unavailable. No uninspected result is used to discharge a proof obligation. This is a focused source comparison, not an exhaustive literature review or a priority claim.